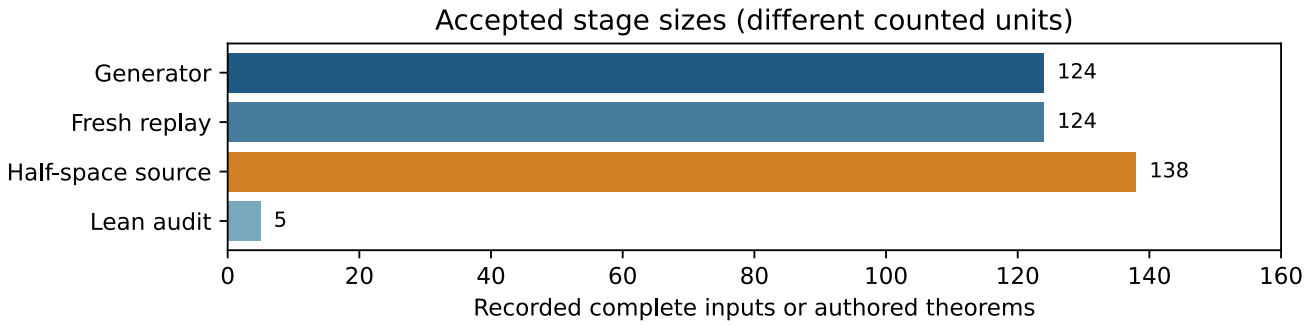
R63: degree 112 source and half-space descent

The actual marked-23 source has degree 112 over F2(x). Its 112 standard monomials and 124 complete source functions have square pairs built from two checked generator formulas and derivative inverses. A fresh source replay binds all 124 saved inputs and roots, both generator bundles, both inverse products, and altered-inverse/root controls.



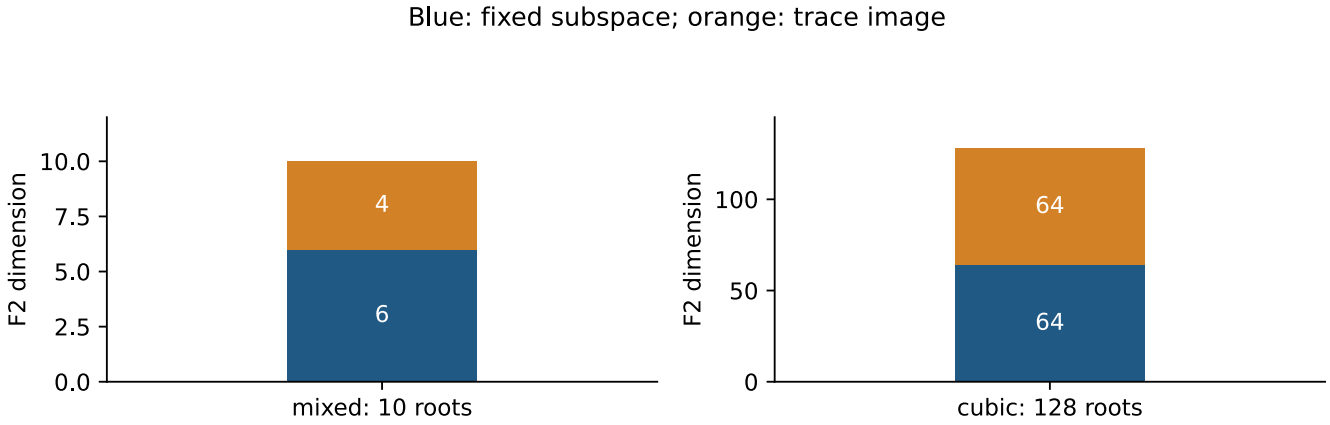
Each bar counts a different exact unit. The two 124-function checks are producer and fresh replay of the same cohort, not independent functions.

Terminal flow

Attempt	Status	Native wall seconds
check112-01	accepted	56
generator112-01	accepted	110
half-descent63-01	rejected	4
half-descent63-02	accepted	32
half-lean63-01	accepted	—

The first half-space attempt is rejected despite a late marker: an unexported ambient constant caused Singular question-mark errors. Version 5 reconstructs the constant from retained R61 matrices; attempt 02 passes. Wall intervals are receipt timing, not a performance comparison.

Accepted half-space restriction



The mixed space has fixed dimension 6 and trace rank 4. The cubic space has fixed dimension 64 and trace rank 64. Both contain an actual function with source involution trace one. The 10 mixed roots lie inside the 128 cubic roots; the two dimensions do not add.

Exact checks and next geometry

The half-space run binds all 138 saved roots to the actual R61 source and verifies $g^2 = f$ for each. It checks 20 mixed and 256 cubic root/square involution identities, all 70 fixed roots, and both trace-one functions in the full source quotient. Each restricted matrix has a checked left inverse, intertwines the ambient action, squares to the identity, and rejects an altered action.

The pinned Lean 4.34.1/mathlib d13f23b audit compiles five authored conditional implications: square-root uniqueness, fixedness of a root over a fixed square, integer pole-cap halving, restriction of an involution, and a trace separator. The audit reports zero failing and zero trusted native decisions. The actual source equations are checked by the native field calculations, and proper pole valuations still need their own construction.

New normalization route

The retained rank-32 trace Gram discriminant predicts a quotient of F2 dimension 160 between the trace-dual lattice and the affine order. The Singh–Swanson descending Frobenius algorithm turns each normalization step into a complete binary kernel on that finite quotient. Equality is an exact affine stopping certificate after the source-field bridge is verified. The old PID variable is z=1/u relative to the actual FIELD61 coordinate. The quadratic equations match under V=z²v; the degree-16 equation and all 32 basis functions require explicit reduction. See R64/NORMALIZATION-LEAD.md for the source-bound algorithm and proof scope.

Proper normalization, branch valuations at zero/one/infinity, complete Riemann–Roch spaces, the level-77 map and order-11 action, quotient saturation, arbitrary reverse divisors, node187, and selected-991 factor action remain separate exact construction tasks.

Reproduction

Five immutable raw attempts; four accepted and one rejected. The 29 historical source byte pins and every terminal receipt are in results.json. Singular used seed 63, one thread, 1200 seconds and 4 GiB virtual memory per run; Lean compile and axiom audit had separate 1200-second caps and the pinned toolchain. Raw archive custody is recorded separately.

Primary normalization source: Singh–Swanson, Theorem 1.1.